MIL-C-24643/22B
3 October 1986
SUPERSEDING
MIL-C-0024643/22A(SH)
8 November 1985
MIL-C-24643/22
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 5000 VOLTS, TYPE LS5KVTSGU
(INCLUDING VARIATION LS5KVTSGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated (see table I for size).
Second - Silicone rubber-glass tape insulation (see table I for diameter). Letter identification code applied by method 5.
Third - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly. Cabling sequence shall be A, B and C.
Fourth - Binder or combination binder and barrier.

Construction variant for type LS5KVTSGU

Fifth - Cross-linked polyolefin jacket. Nominal thickness 0.090 inch. Cable surface marking required.

Construction variant for type LS5KVTSGA

(B) Fifth - Cross-linked polyolefin jacket. Nominal thickness 0.090 inch. Cable surface marking not required.
Sixth - Braided metal armor and paint.

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

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TABLE I. Details.

Military 1/ part no. M24643/22	Size 2/	Conductor size		Type's overall diameter (inches)				Diameter over insulation (nominal) (inches)	Conductor resistance per 1000 feet at 25°C (maximum) (ohms)	Insulation resistance per 1000 feet (minimum) (megohms)	Cold bending mandrel diameter (maximum) (inches)	Accelerated service loading amperes
				LS5KVTSGU		LS5KVTSGA						
		Navy standard	AWG	min	max	min	max					
-01UN	100	----	0(class D)	---	1.74	---	1.79	0.675	0.106	400	22	295
-02UN	150	----	000 (class D)	---	1.95	---	2.00	.769	.0674	350	24	400
-03UN	250	----	250,000 (class C)	---	2.22	---	2.27	.889	.0453	300	28	535
-04UN	350	----	350,000 (class D)	---	2.45	---	2.50	.994	.0321	265	30	665
-05UN	400	400(127)	----	---	2.60	---	2.65	1.054	.0273	250	32	750

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS5KVTSGU and LS5KVTSGA.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
(B) Conductor resistance - ohms/1000 feet, at 25°C, maximum.	(see table I)
Voltage withstand - volts, root mean square, minimum....	13,500
Insulation resistance.....	(see table I)
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
Group B:	
Gas flame - 1 hour.....	No failure
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum.....	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum.....	50
Elongation - percent of unaged, minimum	50
Shrinkage.....	No failure
Heat distortion - percent of unaged, maximum.....	30
Permanence of printing (jacket) - cycles, minimum.....	250
Cable filler removability.....	No failure
Armor (type LS5KVTSGA only) - conformance to material construction and coverage.....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cold working (minus 20 ± 2°C).....	No damage
Gas flame (3 hours).....	No failure
Accelerated service - all sizes (see table I for load current).....	No failure
Aging and compatibility (cable) (125 ± 5°C).....	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum.....	75

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	---
(B) Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:
All types and sizes: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-22)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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